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Assignment-2

Sol-1

(a) The given inequality is not correct

Counter example -

For $G_1 = K_2$ | $\beta_0^{(G)} = 1$ but $w = 2$
$\alpha^{(G)} = 1$

$G_1^c = K_2^c$.

$\Rightarrow \beta_0(G^c) = 0$

$\alpha_0(G^c) = 2$

So $\beta_0(G) + \beta_0(G^c) = 1 + 0 = 1$

$\alpha_0(G) + \alpha_0(G^c) = 1 + 2 = 3$

$\Rightarrow \beta_0(G) + \beta_0(G^c) < \alpha_0(G) + \alpha_0(G^c)$

which is contradicting the inequality given in question.

(b) In this also given inequality is not correct.

Counter example -

$G = K_2$ | $\beta_0(G) = \underline{1}$
but $w(G) = 2$

$\Rightarrow \beta_0(G) < w(G)$ which is a contradiction to inequality $\beta_0(G) \geq w(G)$.

Instead the inequality $B_0(G) \geq w(G) - 1$ holds for every graph.

Proof:

Let K be a clique with $w(G)$ vertices and let C be any vertex cover of G.

Now suppose two vertices $u, v \in V(K)$ are both outside C. Then the edge uv has neither endpoint in C. This contradicts the definition of a vertex cover.

Therefore at most one vertex of K can lie outside C

Hence at least $w(G) - 1$ vertices of K must belong to C

$$\therefore |C| \geq w(G) - 1$$

This is true for every vertex cover, including minimum vertex cover

$$\therefore B_0(G) \geq w(G) - 1$$

(5) Let C be a minimum vertex cover of G.

Therefore $|C| = B_0(G)$

$\therefore$ C is minimum vertex cover, there exists at least one vertex $v \notin C$

Now because C covers every edge, every neighbour of v must belong to C.

$$\therefore \deg(v) \leq |C| \quad - (1)$$

Also $$\delta(G) \leq \deg(v) \quad - (2)$$

Combining (1) and (2), we get:

$$\delta(G) \leq \deg(v) \leq |C| = \beta_0$$

$$\Rightarrow \boxed{\delta(G) \leq \beta_0}$$

Hence proved.

Sol-2 Consider the graph formed by M ∪ N

∵ M and N are both matchings, every vertex is incident to at most one edge of M and at most one edge of N.

∴ Every vertex has degree at most 2 in the subgraph formed by M ∪ N

Moreover, edges in each component must alternate between M and N.

Also Every component of M ∪ N is an alternating path or an alternating cycle.

On an alternating cycle, the numbers of M edge and N edges are equal, otherwise two adjacent will belong to same matching which is a contraction to the fact that the edges in one matching are not adjacent.

On an alternating path, two cases-

Case 1- If end edges belong to different matchings, the two numbers are equal.

Case-2 If end edges belong to same matching then the matching to which they belong will be M and there is exactly one more M-edge than N-edge.

∴ $|M| > |N|$, the total numbers of M edges exceeds the total number of N edges. Therefore, at least one Component must be alternating path with one more M-edge than N-edge.

Let this path be P.

Let

$$|M \cap E(P)| = r+1 \quad , \quad |N \cap E(P)| = r$$

Now define.

$$M' = (M \setminus E(P)) \cup (N \cap E(P))$$

$$N' = (N \setminus E(P)) \cup (M \cap E(P))$$

In other words, swap the membership of every edge along P; edges that belonged to M now belong to N', and edges that belonged to N now belong to M!

Because the edges still alternate along P, both new sets are still matchings.

For M':

$$|M'| = |M| - (r+1) + r$$

$$= \boxed{|M| - 1}$$

For N':

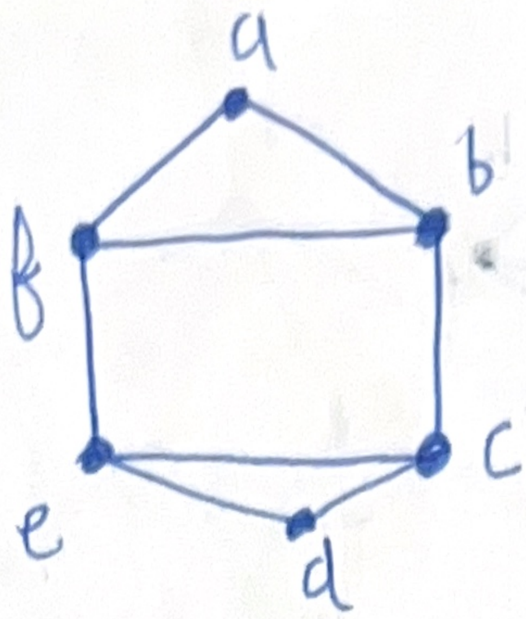
$$|N'| = |N| - r + (r+1)$$

$$= \boxed{|N| + 1}$$

We have only transformed edges between the two matchings. No edge has been added to or removed from their combined set.

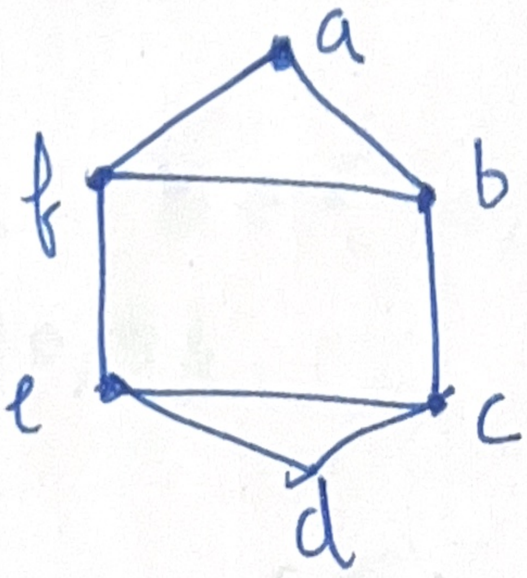
$$\boxed{M' \cup N' = M \cup N}$$

Sol-3

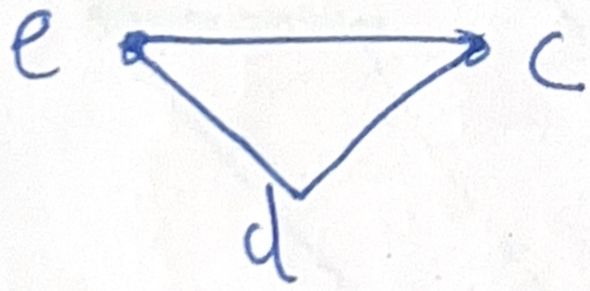


The given graph in the question.

(a) G, $I \leftarrow \phi$



(b) $I \leftarrow \{a\}$, $G_1 = G - \{a\} - \{f, b\}$



G_1

(C) $I \leftarrow \{a, d\}$

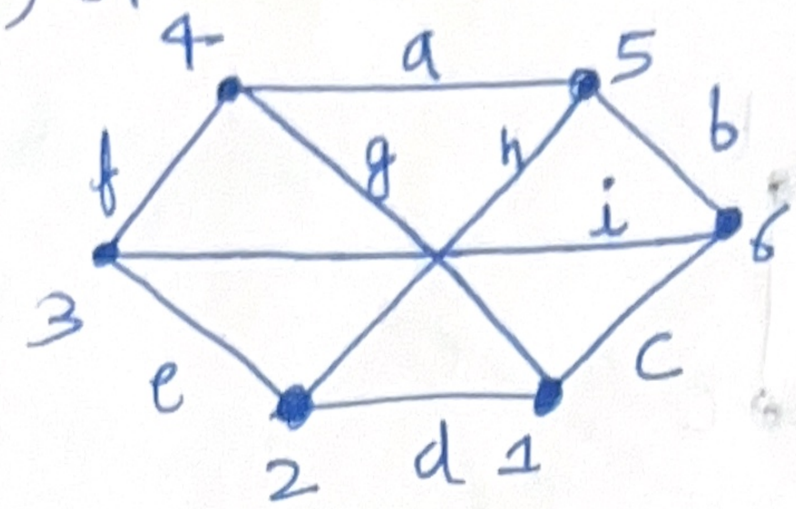
$G_2 = G_1 - \{e, c, d\}$

Now G_2 graph with no vertices.

$\therefore I = \{a, d\}$

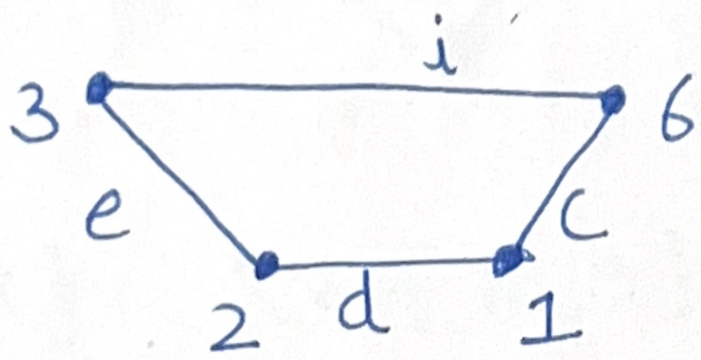
$|I| = 2$

Sol - 4. (a) G_1



(b) $M \leftarrow \{a\}$ $K \leftarrow \{4, 5\}$

$G_1 = G - \{4, 5\}$



(c) $M \leftarrow \{a, i\}$ $K \leftarrow \{4, 5, 6, 3\}$

$G_2 = G_1 - \{3, 6\}$

(d) $M \leftarrow \{a, i, d\}$ $K \leftarrow \{1, 2, 3, 4, 5, 6\}$

$G_3 = G_2 - \{1, 2\}$

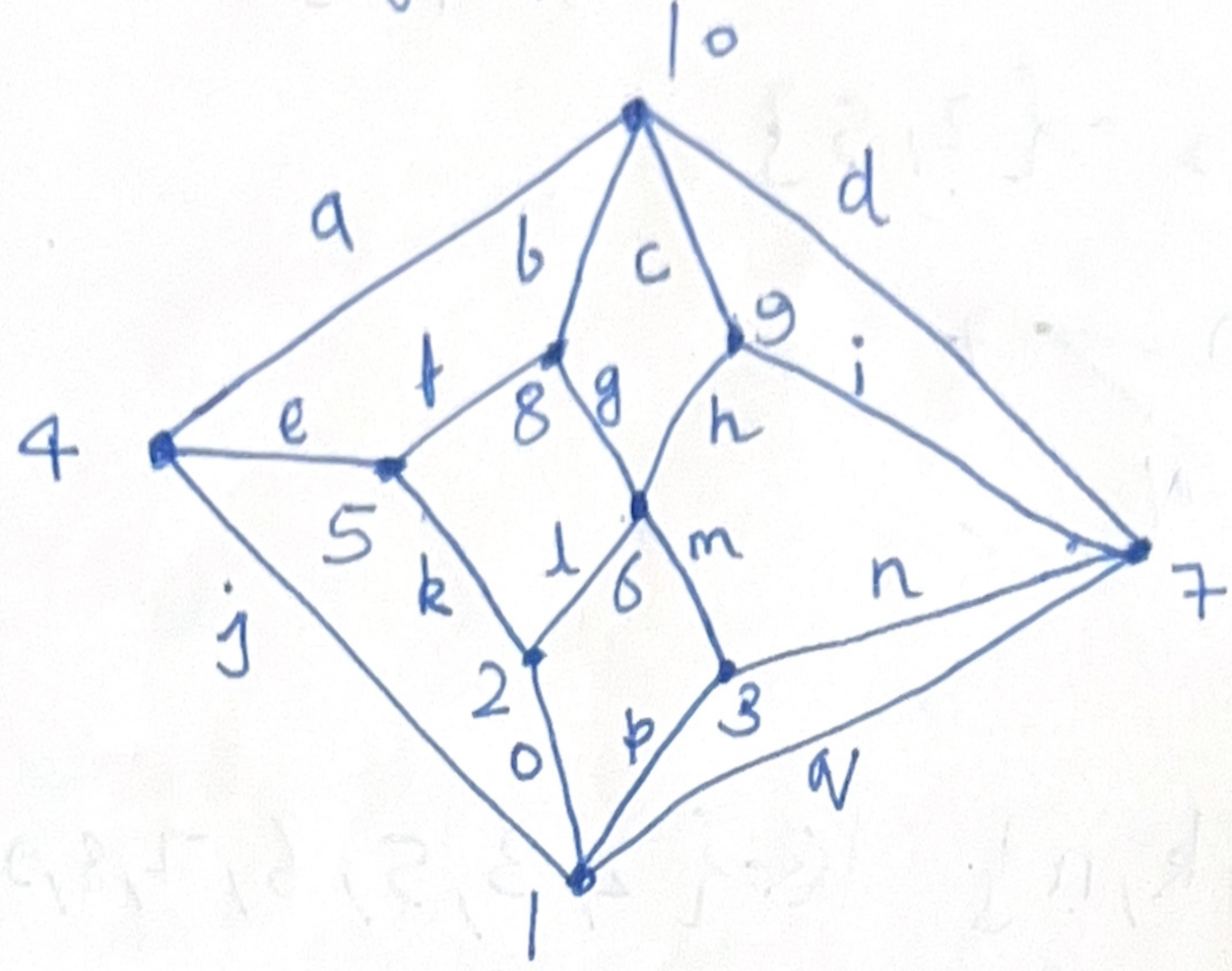
G_3 is graph left with no vertices.

So $M = \{a, i, d\}$

$K = \{1, 2, 3, 4, 5, 6\}$

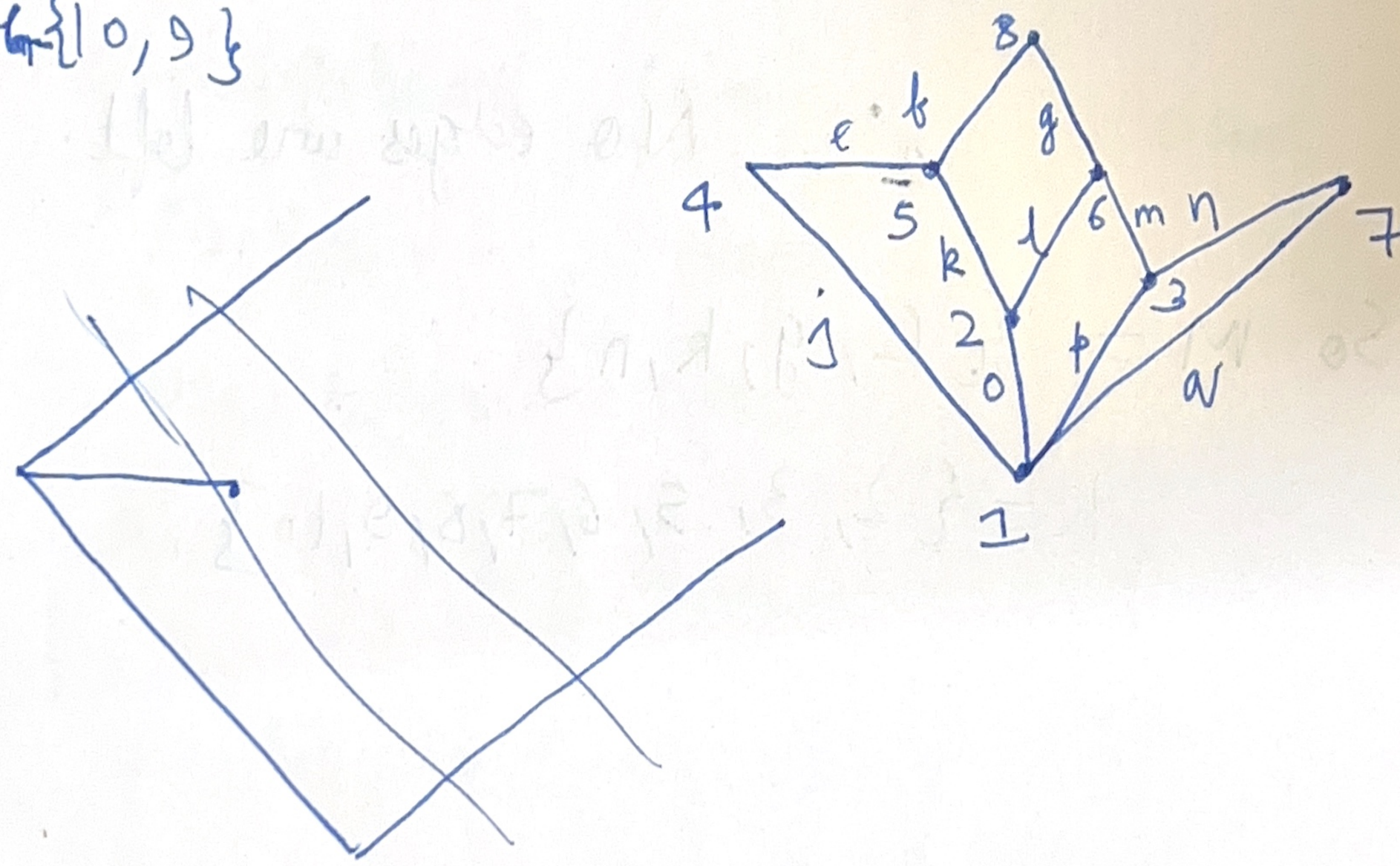
For second graph

(a) G



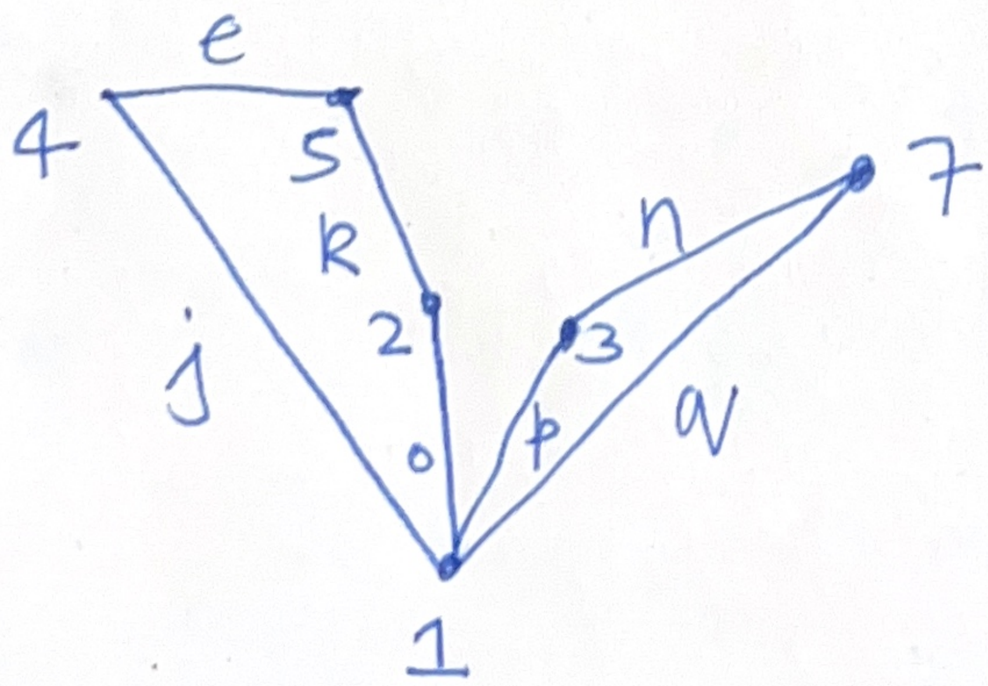
(b) $M \leftarrow \{c\}$ $K \leftarrow \{10, 9\}$

$G_1 = G - \{10, 9\}$



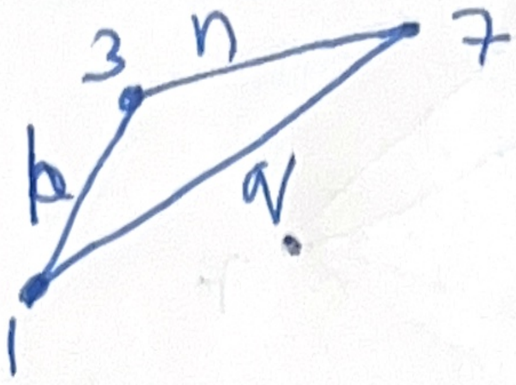
(c) $M \leftarrow \{c, g\}$ $K \leftarrow \{6, 8, 9, 10\}$

$G_2 = G_1 - \{6, 8\}$



(d) $M \leftarrow \{c, g, k\}$ $K \leftarrow \{2, 5, 6, 8, 9, 10\}$

$G_3 = G_2 - \{2, 5\}$



(e) $m \leftarrow \{c, g, k, n\}$ $K \leftarrow \{2, 3, 5, 6, 7, 8, 9, 10\}$

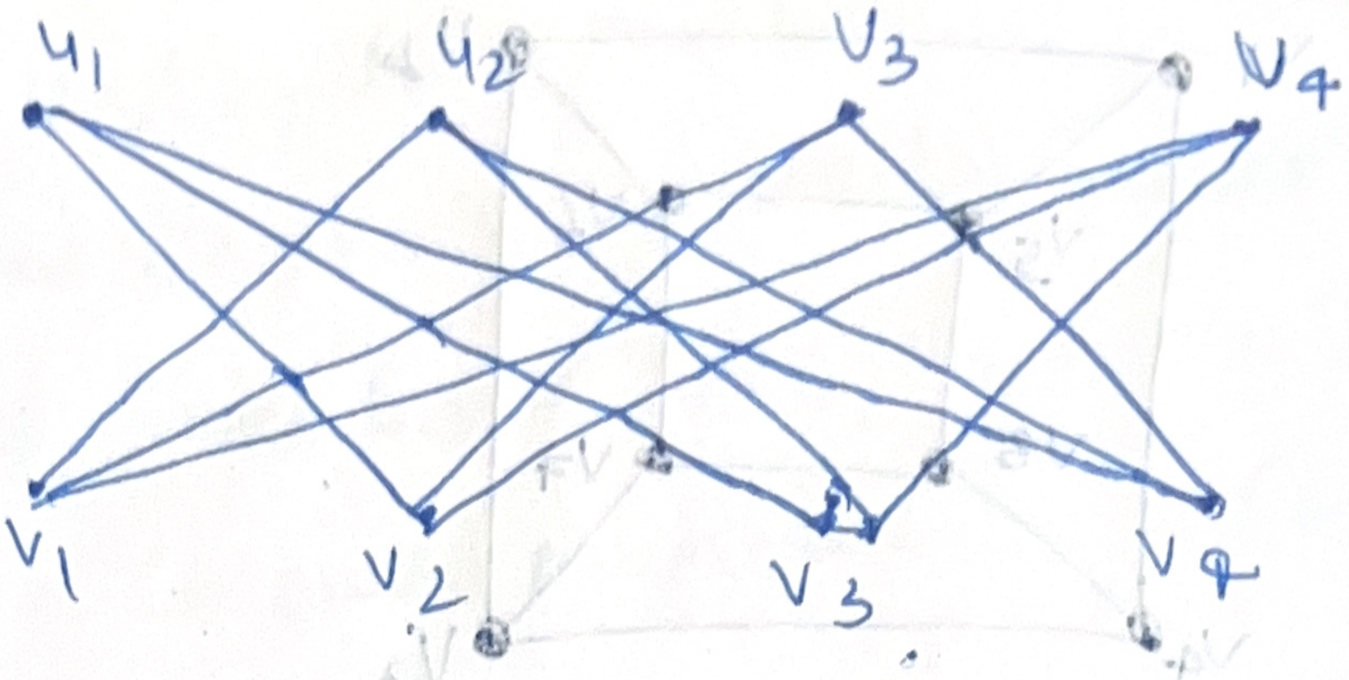
$G_4 = G_{43} - \{3, 7\}$

1 No edges are left.

So $M = \{c, g, k, n\}$

$K = \{2, 3, 5, 6, 7, 8, 9, 10\}$.

Sol-5

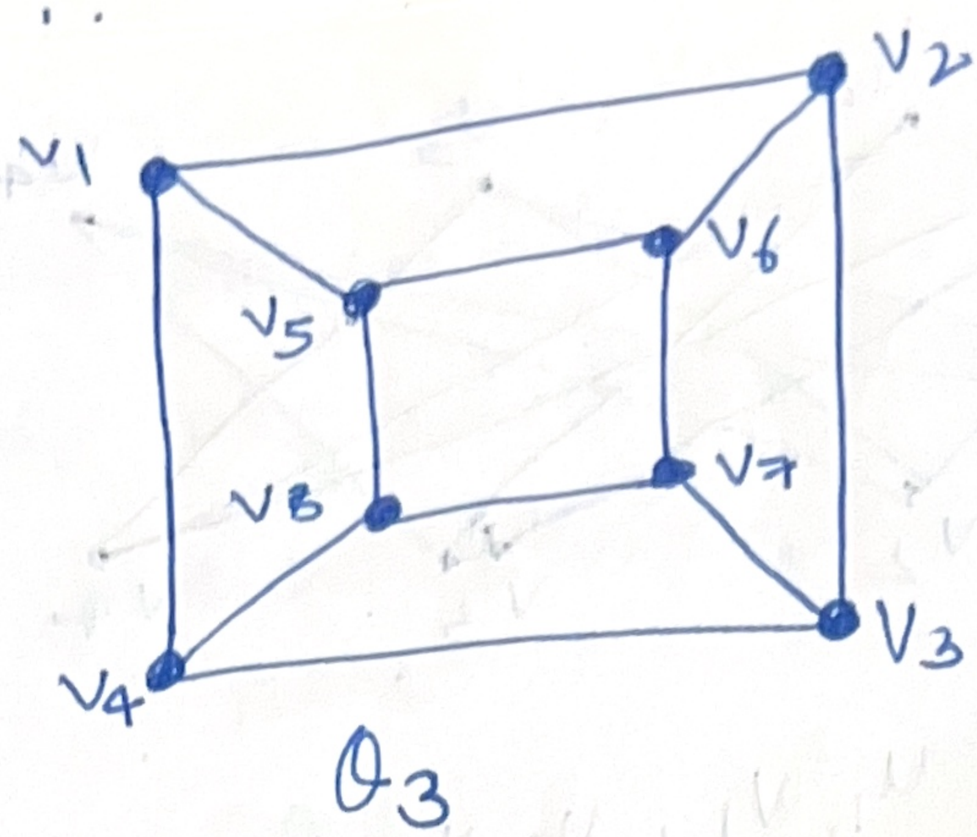


Order Take $S = \{u_1, v_1, v_2, v_2, u_3, v_3, v_4, v_4\}$.

In the greedy algorithm, each vertex receives the smallest colour not used on its already coloured neighbours.

Vertex	Neighbours	Coloured Neighbours	Colour
u_1	v_2, v_3, v_4	none.	1
v_1	u_2, u_3, u_4	none	1
u_2	v_1, v_3, v_4	$v_1(1)$	2
v_2	u_1, u_3, u_4	$u_1(1)$	2
u_3	u_1, v_2, v_4	$v_1(1), v_2(2)$	3
v_3	u_1, u_2, u_4	$u_1(1), u_2(2)$	3
u_4	v_1, v_2, v_3	$v_1(1), v_2(2), v_3(3)$	4
v_4	u_1, u_2, u_3	$u_1(1), v_2(2), v_3(3)$	4

Sol-6



Q_3

It is bipartite with parts $X = \{v_1, v_3, v_6, v_8\}$ and $Y = \{v_2, v_4, v_5, v_7\}$

In the greedy algorithm, each vertex receives the smallest colour not used on its earlier-coloured neighbours.

(a) 2 colours. Take $S_1 = (v_1, v_3, v_6, v_8, v_2, v_5, v_7)$

That is, all of X first and then all of Y.

- The vertices of X are pairwise non adjacent, so each gets colour 1.
- Each vertex of Y has a neighbour in X coloured 1 and no neighbour in Y, so each gets colour 2.

Only colours 1 and 2 are used.

(b) 3 colours Take $S_2 = (v_1, v_7, v_2, v_6, v_3, v_4, v_5, v_8)$

Vertex	Coloured neighbours	Color
v_1	none	1
v_7	none (v_7 is not adjacent to v_1)	1
v_2	v_1(1)	2
v_6	v_2(2), v_7(1)	3
v_3	v_2(2), v_7(1)	3
v_4	v_1(1), v_3(3)	2
v_5	v_1(1), v_6(3)	2

	...	3
V_8	$V_5(2), V_7(1), V_4(2)$	

The maximum color is 3, so exactly 3 colors are used.

(c) 4 colours, Take $S_3 = (V_2, V_8, V_4, V_6, V_5, V_1, V_3, V_7)$

Vertex	Coloured neighbours	Color
V_2	none	1
V_8	none (V_8 is not adjacent to V_2)	1
V_4	$V_8(1)$	2
V_6	$V_2(1)$	2
V_5	$V_8(1), V_6(2)$	3
V_1	$V_2(1), V_4(2), V_5(3)$	4
V_3	$V_2(1), V_4(2)$	3
V_7	$V_6(2), V_8(1), V_3(3)$	4

Here 4 colours are used.

Sol - 7

(a) Let $d = \Delta(T_k)$ $q = 2k+1$

Finding the maximum degree

A vertex x in the tree is connected to all q cycle vertices, in addition to its original tree neighbours.

A cycle vertex is connected to all k tree vertices and its two cycle neighbours.

$\therefore \Delta G = \max\{q+d, k+2\}$.

$\therefore q = 2k+1$.

$2k+1+d \geq k+2$

$\therefore \Delta(G) = 2k+1+d$

Let the tree vertices be

$t_0, t_1, \cdots t_{k-1}$ and cycle vertices be $c_0, c_1 \cdots c_k$

The cross edges form $K_{k,2k+1}$. Use $2k+1$ colours.

For each $i \in \{0,1 \cdots 2k\}$ define the matching.

$$M_i = \{ t_j c_{(i+j) \bmod (2k+1)} : 0 \le j \le k-1 \}.$$

Each matching contains exactly one edge incident with every vertex.

Moreover, these matchings partition all the cross edges; every cross-edge $t_j c_i$ appears exactly once in the matching with

$$i \equiv l - j \pmod{2k+1}$$

Notice that M_i uses the cycle vertices $c_i, c_{i+1} \cdots, c_{i+k-1}$ with indices taken modulo $2k+1$.

Consequently, the vertices.

$c_{i+k}, c_{i+k+1}, \cdots, c_{i+2k}$ are not incident with any cross edge of colour i.

In particular, both endpoints of the cycle edge

$c_{i+k} c_{i+k+1}$ are free of colour i.

$\therefore$ Add that cycle edge to M_i, As i runs through all $2k+1$ values, these are precisely all the edges of the cycle. So the cross edges and cycle edges have been properly colored using $2k+1$ colours.

Now, the tree has maximum degree d and it is bipartite

$\therefore$ Its internal edges can be coloured by using additional d colours.

Total no. of colors $= 2k+1 + d = \Delta(G_1)$

$$\boxed{\chi_1(T_k + C_{2k+1}) = 2k+1 + \Delta(T_k)}$$

Each matching.

(b) $C_5 + C_5 = G_1$

This graph is the join of two 5-cycles

Let the vertices of the two cycles be

$A = \{a_1, a_2, a_3, a_4, a_5\}$

$B = \{b_1, b_2, b_3, b_4, b_5\}$.

Each vertex has 2 neighbours in its own cycle and 5 neighbours in its own cycle.

$\therefore \Delta(G_1) = 2 + 5 = 7$.

Hence $\chi_1(G_1) \geq 7$

We will partition all edges into seven matchings

Use the following seven colour classes.

$M_1 = \{a_1a_2, a_3a_4, a_5b_1, b_2b_3, b_4b_5\}$

$M_2 = \{a_1a_5, a_2a_3, a_4b_2, b_1b_5, b_3b_4\}$

$M_3 = \{a_1b_3, a_2b_4, a_3b_5, a_4a_5, b_1b_2\}$

$M_4 = \{a_1b_1, a_2b_2, a_3b_3, a_4b_4, a_5b_5\}$

$M_5 = \{a_1b_2, a_2b_1, a_3b_4, a_4b_5, a_5b_3\}$

$M_6 = \{a_1b_4, a_2b_5, a_3b_1, a_4b_3, a_5b_2\}$

$M_7 = \{a_1b_5, a_2b_3, a_3b_2, a_4b_1, a_5b_4\}$.

Thus they form a proper 7-edge colouring.

Therefore $\boxed{\chi_1(G_1) = 7}$

Each matching contains all vertices and hence

(c) Without loss of generality, assume

$s \leq t$.

Let the two cycles $A = C_{2s}$ and $B = C_{2t}$.

A vertex in A has 2 neighbours in A and 2t neighbours in B. Its degree is $2+2t$.

Similarly, a vertex in B has degree $2+2s$

$\therefore$ $\boxed{\Delta(G) = 2t+2}$

Case -1 $s = t$.

This cross-edges form $K_{2s,2s}$

Since this graph is bipartite and 2s regular, its edges can be partitioned into 2s perfect matchings.

Use 2s colours for the cross edges.

Now both cycles have even length. An even cycle can be properly edge-coloured using two colours alternately.

Use two additional colours for the internal edges of both cycles.

$\therefore$ Total no. of colors $= 2s+2 = 2t+2$.

Case 2: $s < t$

Let $A = \{a_0, \ldots a_{2s-1}\}$

$B = \{b_0, \ldots b_{2t-1}\}$

The cross edges form $K_{2s,2t}$

We use 2t colours indexed by

$i = 0, 1, \ldots, 2t-1$

For each i, construct the matching

$M_i = \{a_j b_{(i+j) \bmod 2t} : 0 \leq j \leq 2s-1\}$.

Each matching saturates all vertices of A and these $2t$ matchings partition cross edges.

Matching M_i uses the vertices.

$$b_i, b_{i+1}, \ldots, b_{i+2s-1}$$

Therefore, it does not use.

$$b_{i+2s}, b_{i+2s+1}, \ldots, b_{i+2t-1}$$

$\therefore s<t$, the vertices b_{i+2s} and b_{i+2s+1} are both unused in M_i

So we can add the cycle edge

$b_{i+2s} b_{i+2s+1}$ to M_i, giving it the same colour i

As i ranges over all $2t$ values, we colour every edge of cycle B exactly once.

Finally, the remaining cycle $A = C_{2s}$ is even, so its edges can be coloured alternately using two new colours

This, the total no. of colors is $2t+2$

$\Delta(G) = 2t+2$, we conclude.

$$\boxed{\chi_1(C_{2s} + C_{2t}) = 2\max(s,t)+2}$$

Sol-8 Form the bipartite graph $G_1[A, D]$ where $A = \{A_1, \ldots, A_7\}$ and $D = \{$Chemistry, English, French, Geography, History, Mathematics, Physics$\}$. A feasible staffing is a matching, and the maximum number of teachers employed equals the matching number of G_1. Since $|D| = 7$, the matching number is at most 7.

Claim: The maximum is 7, so every vacancy can be filled.

Justification. The following matching saturates both sides:

- A_1 to Mathematics.
- A_6 to Physics.
- A_5 to Chemistory.
- A_2 to English.
- A_7 to Geograph
- A_3 to French
- A_4 to History.

Sol-9 Assumption- Considering the graph to be loopless.

By handshaking lemma

$$\sum_{v \in V(G)} \deg(v) = 2|E(G)|$$

~~$\therefore$ Every vertex has degree at most $\Delta(G)$.~~

~~$2|E(G)| \leq n\Delta(G)$~~

Suppose G has a $\Delta(G)$ edge colouring.

Since n is odd, the maximum size of matching is $\frac{n-1}{2}$

Each colour class is matching. Therefore, the maximum number of edges that can be covered by $\Delta(G)$ colours is $(\Delta G)\frac{(n-1)}{2} \Rightarrow E(G) \leq (\Delta G)\frac{(n-1)}{2}$

But, the assumed number of edges is greater than this $[2|E(G)| > (n-1)\Delta(G)]$

Hence there is contradiction.

Thus $\chi_1(G) > \Delta(G)$

∴ Since G is simple, Vizing's theorem gives

$\chi_1(G) \leq \Delta(G) + 1$

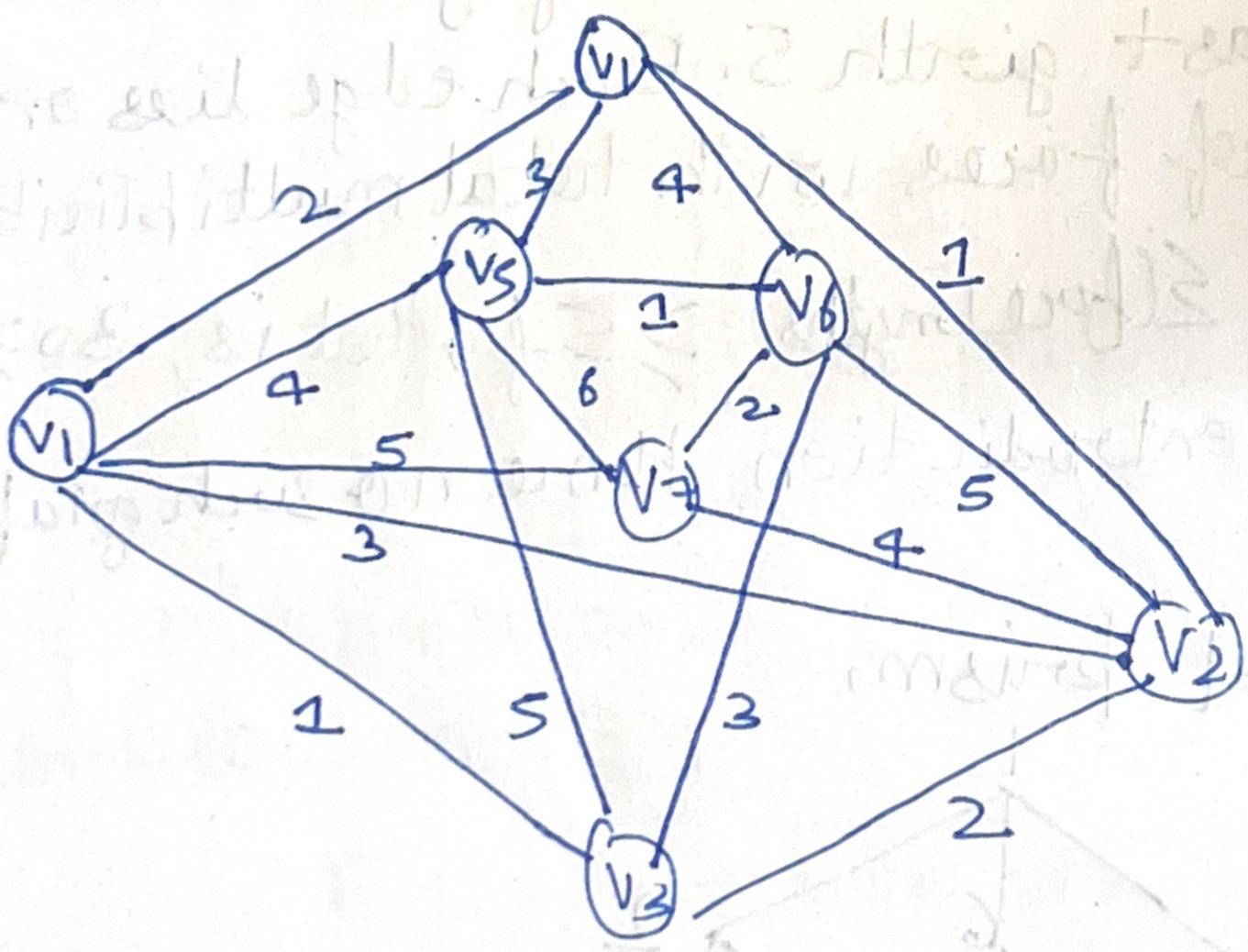
Combining

$$\boxed{\chi_1(G) = \Delta(G) + 1}$$

In the given graph, $n = 7$ (odd) and $\Delta = 5$

Then $2m = 32 > 30 = (n-1)\Delta$, so the statement applies and $\chi_1(G) = \Delta + 1 = 6$.

A proper 6-edge colouring. The numbers on the edges in drawing are the colors



Colour	Edges
1	v_1v_2, v_3v_4, v_5v_6
2	v_1v_4, v_2v_3, v_6v_7
3	v_1v_5, v_2v_4, v_3v_6
4	v_1v_6, v_2v_7, v_4v_5
5	v_2v_6, v_3v_5, v_4v_7
6	v_5v_7

Sol-10(a)

Suppose G is a simple 3 regular Planar graph with $n=10$ and girth 5. Then $m = \frac{3n}{2} = 15$.

G is connected. Fix a vertex v. Its 3 neighbours are distinct and each has 2 further neighbours.

These $1+3+6=10$ vertices are pairwise distinct, because a coincidence would create a cycle of length at most 4. So the component of V already contains all 10 vertices.

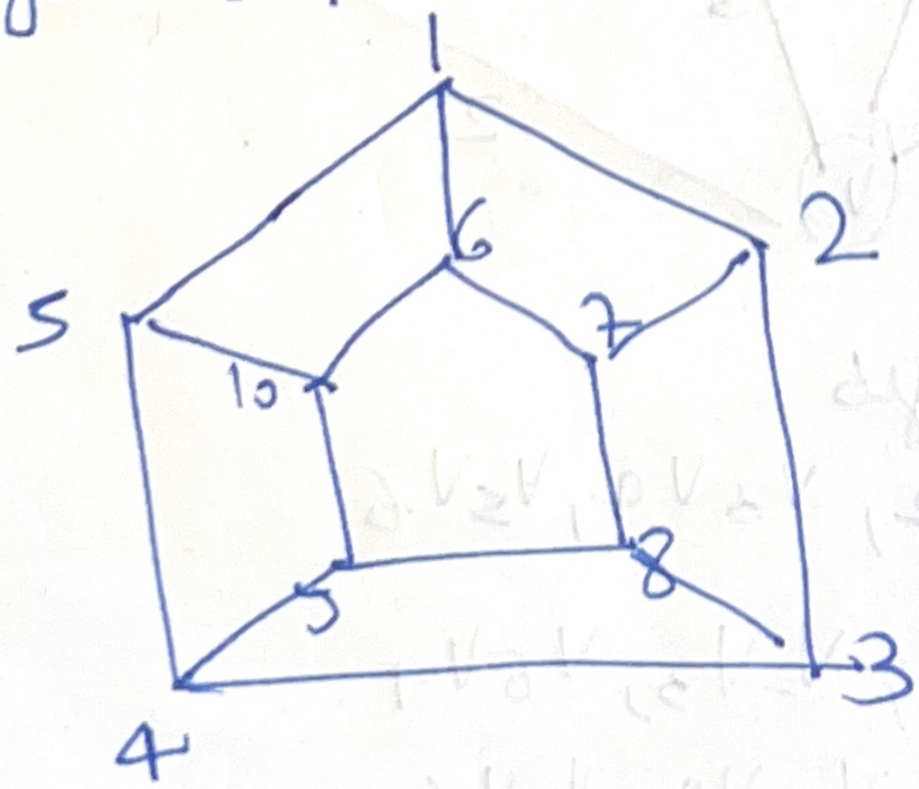
Fix a plane embedding with f faces. Euler's formula gives $n-m+f=2$, so $f=2-10+15=7$.

G contains cycles and every face boundary has length at least girth 5. Each edge lies on the boundaries of faces with total multiplicity 2, so

$2m = \Sigma(\text{face lengths}) \geq 5f$, that is, $30 \geq 35$.

This is a contradiction. Hence no such graph exists.

(b) Pentagonal prism.



Sol-11

We are given the graph is 3 regular

Every vertex has degree 3

By Handshaking Lemma,

$$\therefore \sum deg(V) = 2m$$

$$\Rightarrow 3n = 2m \Rightarrow \boxed{m = \frac{3n}{2}} \quad -(1)$$

There are p pentagons, each of degree 5, h hexagons each of degree 6

$\therefore 5p + 6h$ is the sum of degrees of all faces.

In a plane graph, every edge belongs to exactly two faces, so $\sum deg(faces) = 2m$

Thus $5p + 6h = 2m$

From (1)

$$\boxed{5p + 6h = 3n} \quad -(2)$$

For a connected plane graph, by Euler's formula $n - m + f = 2$ and we have no. of faces $f = p + h$.

So, $n - m + p + h = 2$

Substitute $m = \frac{3n}{2}$

$$n - \frac{3n}{2} + p + h = 2$$

Thus, $$\frac{-n}{2} + p + h = 2$$

$$\Rightarrow \boxed{n = 2p + 2h - 4} \quad -(3)$$

From combining Eqn. (2) and (3), we get:

$$3(2p + 2h - 4) = 5p + 6h$$

$$\Rightarrow \boxed{p = 12}$$

Now substitute $p=12$ into (3)

$$n = 2(12) + 2h - 4$$

Hence $\boxed{n = 20 + 2h}$

Sol-12

Let G_1 be a simple planar graph with $n \geq 11$ vertices

We need to prove

$\bar{G}_1$ is nonplanar

They key fact is: A simple planar graph with $n \geq 3$ vertices has at most $3n-6$ edges

$$|E(G_1)| \leq 3n-6$$

Now G_1 and its complement together contain all possible edges of K_n.

$$|E(G_1^c)| = \binom{n}{2} - |E(G_1)| \geq \frac{n(n-1)}{2} - (3n-6)$$

$$= \frac{(n-3)(n-4)}{2}$$

For $n \geq 11$

$$\frac{(n-3)(n-4)}{2} \geq 3n-6$$

Hence $\bar{G}_1$ has more than $3n-6$ edges which is impossible for a simple planar graph. Thus $\bar{G}_1$ is nonplanar.

Now we need an 8 vertex graph G_1 such that $G_1 \cong G_1^c$ and G_1 is planar

Since a self complementary graph on 8 vertices must have $\frac{1}{2}\binom{8}{2} = \frac{1}{2} \times 28 =$ 14 edges

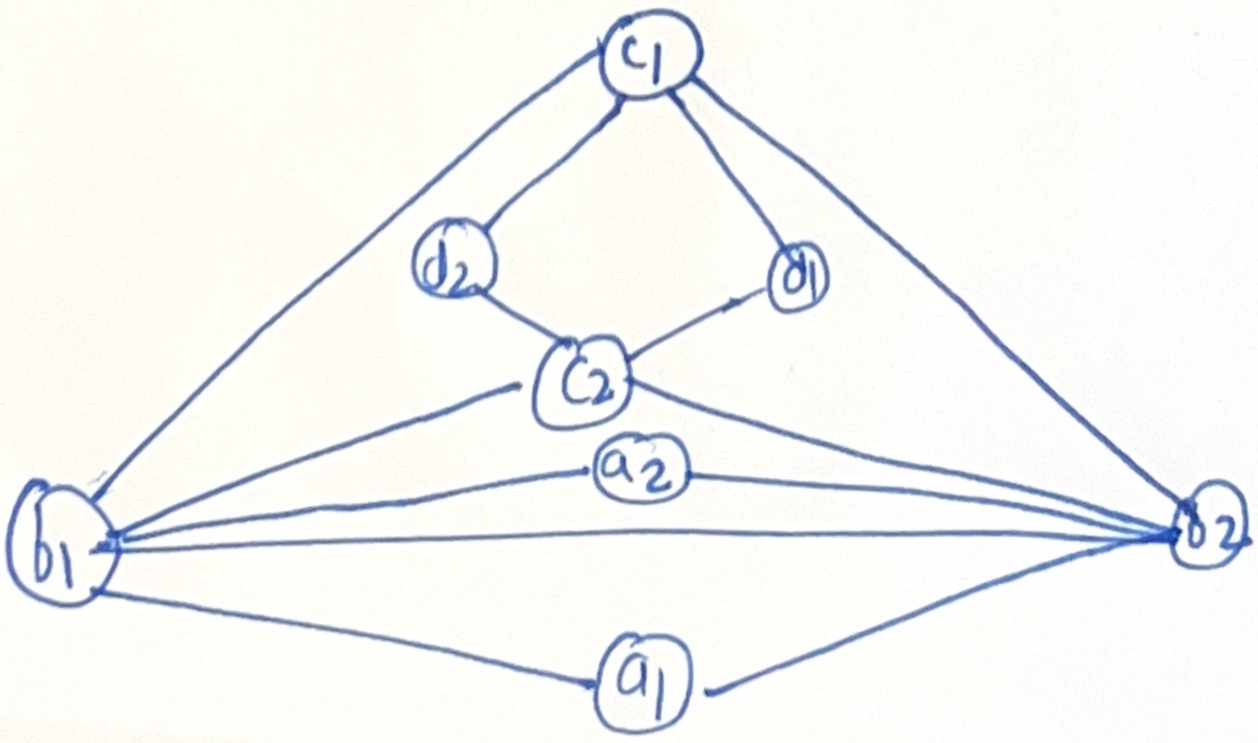
Partition 8 vertices into 4 pairs $V_1 = \{a_1, a_2\}, V_2 = \{b_1, b_2\}$, $V_3 = \{c_1, c_2\}$ and $V_4 = \{d_1, d_2\}$

The graph G_1 has these edges.

- $b_1 b_2$ and $c_1 c_2$, so V_2 and V_3 each induce K_2
- all edges between V_1 and V_2, between V_2 and V_3 and between V_3 and V_4.

V_1 and V_4 are independent and there are no other edges. The edge count is $1 + 1 + 4 + 4 + 4 = 14 = C(8,2)/2$ which is the count a self-complementary graph on 8 vertices must have.

In G_1^c, the sets V_1 and V_4 become K_2, the sets V_2 and V_3 become independent, and the complete joins are exactly V_3 to V_1, V_1 to V_4, and V_4 to V_2. So G_1^c is the same structure along the path V_3, V_1, V_4, V_2.



The drawing has 8 vertices and 14 edges and no crossings. It has 8 faces, consistent with Euler's formula $8 - 14 + f = 2$ and $14 \leq 3 \cdot 8 - 6 = 18$ as the planar graph requires.